WORKED SOLUTION

STEP III 2019, Question 3

Pure • Invariant Lines of Matrices

STEP III 2019, Question 3

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

A point (x, y) is invariant exactly when $(\mathbf{A} - \mathbf{I})$ sends it to zero, which gives two linear equations. Everything in parts (i) and (ii) comes from asking when those two equations describe the same line, and the condition for that is $(a-1)(d-1) = bc$, which says $\det(\mathbf{A} - \mathbf{I}) = 0$. In part (iii) we substitute a general point of the line into the transformation and compare coefficients.

Solution

Part (i)

The point (x, y) is invariant when $ax + by = x$ and $cx + dy = y$, that is,

$$(a-1)x = -by, \tag{1}$$
$$(d-1)y = -cx. \tag{2}$$

Multiplying (1) and (2) together gives $(a-1)(d-1)xy = bc\,xy$, so

$$((a-1)(d-1) - bc)xy = 0$$

for every point (x, y) of L_1, as required.

For the second result, we want a point of L_1 with $xy \neq 0$, so that we can divide by xy.

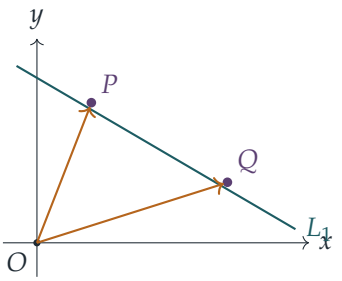
? **Ask yourself** What if every point of L_1 has $x = 0$ or $y = 0$?

The set $xy = 0$ is just the two coordinate axes. A line lying inside the union of the axes has to be one of them, so unless L_1 is an axis it contains a point with $xy \neq 0$. Dividing by xy then gives $(a-1)(d-1) = bc$.

That leaves the axes. If L_1 is the y-axis, every $(0, y)$ is invariant, and (1) and (2) give $by = 0$ and $(d-1)y = 0$ for all y. So $b = 0$ and $d = 1$, and both sides of

$(a-1)(d-1) = bc$ are 0. If L_1 is the x-axis then in the same way $c = 0$ and $a = 1$, and again both sides are 0. So $(a-1)(d-1) = bc$ in every case.

Now suppose L_1 misses the origin. Take two distinct points P and Q on L_1, with position vectors $\mathbf{p}$ and $\mathbf{q}$.



Since O, P and Q do not lie on one line, $\mathbf{p}$ and $\mathbf{q}$ are not parallel, so every vector can be written as $\mathbf{v} = \lambda\mathbf{p} + \mu\mathbf{q}$. By linearity,

$$\mathbf{Av} = \lambda\mathbf{Ap} + \mu\mathbf{Aq} = \lambda\mathbf{p} + \mu\mathbf{q} = \mathbf{v},$$

so every point of the plane is invariant and $\mathbf{A} = \mathbf{I}$. (Then $a = d = 1$ and $b = c = 0$, which agrees with $(a-1)(d-1) = bc$.)

Part (ii)

Suppose $(a-1)(d-1) = bc$. We need a line on which both (1) and (2) hold.

Case $b \neq 0$**.** Take the line $(a-1)x + by = 0$, which is a genuine line through the origin because $b \neq 0$. Every point on it satisfies (1). On this line $y = -\frac{(a-1)x}{b}$, so

$$cx + (d-1)y = cx - \frac{(a-1)(d-1)}{b}\,x = cx - \frac{bc}{b}\,x = 0,$$

and (2) holds too. So $(a-1)x + by = 0$ is a line of invariant points.

Case $b = 0$**.** Now the condition becomes $(a-1)(d-1) = 0$, so $a = 1$ or $d = 1$.

- If $a \neq 1$, then $d = 1$. Equation (1) reads $(a-1)x = 0$, so $x = 0$, and (2) reads $cx = 0$, which holds when $x = 0$. The line of invariant points is the y-axis, $x = 0$.
- If $a = 1$, then (1) holds for every point, and we only need (2): $cx + (d-1)y = 0$.
  - If c and $d-1$ are not both zero, this is a line through the origin, and it is a line of invariant points.
  - If $c = 0$ and $d = 1$, then $\mathbf{A} = \mathbf{I}$, every point is invariant and every line is a line of invariant points.

So whenever $(a-1)(d-1) = bc$ there is a line of invariant points.

Part (iii)

Write L_2 as $y = mx + k$ with $k \neq 0$, since L_2 does not pass through the origin. A general point of L_2 is $(x,\ mx + k)$, and its image is

$$(X, Y) = \big((a + bm)x + bk,\ (c + dm)x + dk\big).$$

The line is invariant exactly when $Y = mX + k$ for every real x:

$$(c + dm)x + dk = m(a + bm)x + mbk + k.$$

Both sides are linear in x and agree for all x, so the coefficients must match:

$$\text{constant terms:} \quad dk = mbk + k, \tag{3}$$

$$\text{coefficients of } x: \quad c + dm = am + bm^2. \tag{4}$$

Ask yourself Where do we use the fact that L_2 misses the origin?

We use it in (3): because $k \neq 0$ we may divide by k, which gives $d - 1 = mb$. Substituting $d = 1 + mb$ into (4),

$$c = am + bm^2 - dm = am + bm^2 - m - bm^2 = m(a - 1).$$

Therefore

$$bc = bm(a - 1) = (d - 1)(a - 1),$$

which is the required result. (If $k = 0$, equation (3) gives nothing, which is why a line through the origin can be invariant without the condition. A rotation through π is an example: it maps every line through O to itself, but $(a - 1)(d - 1) = 4 \neq 0 = bc$.)

Checks

- Try $\mathbf{A} = \begin{pmatrix} 3 & 2 \\ -1 & 0 \end{pmatrix}$: here $(a - 1)(d - 1) = -2 = bc$. Part (ii) predicts the invariant line $2x + 2y = 0$, that is $y = -x$, and indeed $\mathbf{A}(1, -1)^{\mathrm{T}} = (1, -1)^{\mathrm{T}}$.
- For part (iii), take $m = 2$, $k = 1$, $a = 3$, $b = 1$. Then $d = 1 + mb = 3$ and $c = m(a - 1) = 4$. The point $(1, 3)$ on $y = 2x + 1$ maps to $(6, 13)$, and $13 = 2 \times 6 + 1$. Also $(a - 1)(d - 1) = 4 = bc$.
- Thousands of random matrices were confirmed by computer: every invariant line $y = mx + k$ with $k \neq 0$ found had $d - 1 = mb$, $c = m(a - 1)$ and $(a - 1)(d - 1) = bc$, and every line of fixed points missing the origin came from $\mathbf{A} = \mathbf{I}$.

Common slips

- Dividing by xy in part (i) without dealing with the case where L_1 is a coordinate axis.

- Missing a sub-case in part (ii) when $b = 0$, especially $\mathbf{A} = \mathbf{I}$, where every line works.
- In part (iii), confusing an *invariant line* (points stay on the line) with a *line of invariant points* (points stay put), and so setting $X = x$.

Beyond STEP

The condition $(a-1)(d-1) = bc$ is $\det(\mathbf{A} - \mathbf{I}) = 0$, which says that 1 is an eigenvalue of $\mathbf{A}$. A line of invariant points through the origin is then the eigenspace for that eigenvalue. Part (iii) shows that an invariant line missing the origin also forces the eigenvalue 1. The relations $c = m(a-1)$ and $d - 1 = mb$ say that the second row of $\mathbf{A} - \mathbf{I}$ is m times the first, so $\mathbf{A} - \mathbf{I}$ has rank at most 1 and moves every point parallel to L_2. Eigenvalues, eigenvectors and the geometry of linear maps are covered in Strang, *Introduction to Linear Algebra*.

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